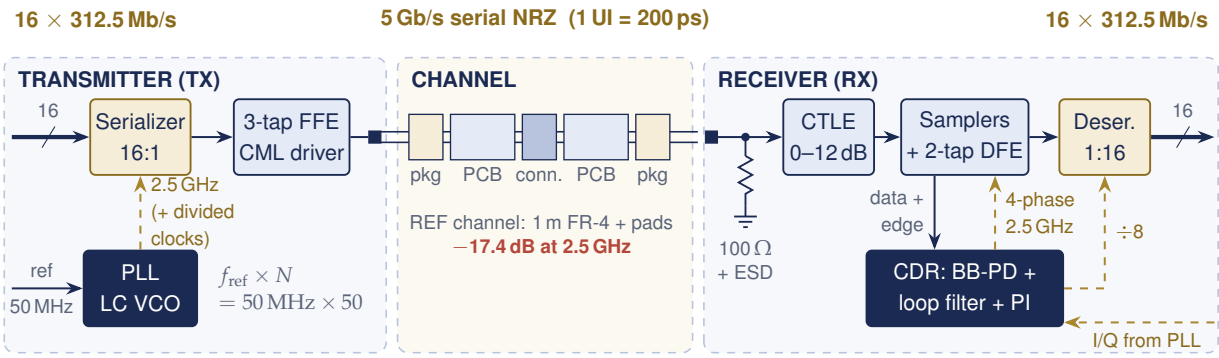

16 × 312.5 Mb/s
5 Gb/s serial NRZ (1 UI = 200 ps)
16 × 312.5 Mb/s
TRANSMITTER (TX)
16
Serializer 16:1
3-tap FFE CML driver
2.5 GHz (+ divided clocks)
ref 50 MHz
PLL LC VCO
$f_{ref} \times N = 50\,MHz \times 50$
CHANNEL
pkg
PCB
conn.
PCB
pkg
REF channel: 1 m FR-4 + pads
−17.4 dB at 2.5 GHz
RECEIVER (RX)
100 Ω + ESD
CTLE 0–12 dB
Samplers + 2-tap DFE
Deser. 1:16
16
data + edge
4-phase 2.5 GHz
÷8
CDR: BB-PD + loop filter + PI
I/Q from PLL